

Assignment 13.2

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Task 1 – Refactoring (Eliminating Repetitive Logic)

```
def rectangle_properties(length, width):  
    """  
    Calculate and print area and perimeter of a rectangle.  
    """  
    area = length * width  
    perimeter = 2 * (length + width)  
  
    print("Area:", area)  
    print("Perimeter:", perimeter)  
  
rectangle_properties(8, 4)  
rectangle_properties(12, 6)
```

Output

```
Area: 32  
Perimeter: 24  
Area: 72  
Perimeter: 36
```

Task 2 – Refactoring (Modularizing Inline Calculations)

```
def calculate_final_amount(amount):  
    """  
    Calculate final amount after applying 10% discount.  
    """  
    discount = amount * 0.10  
    final_amount = amount - discount
```

```
print("Final Amount:", final_amount)
```

```
calculate_final_amount(1000)
```

```
calculate_final_amount(2000)
```

Output

Final Amount: 900.0

Final Amount: 1800.0

Task 3 – Refactoring (Improving Loop and Conditional Performance)

```
students = ["Anil", "Bhavya", "Charan", "Divya"]
```

```
check_list = ["Charan", "Esha", "Bhavya"]
```

```
# Convert list to set for faster lookup
```

```
student_set = set(students)
```

```
for name in check_list:
```

```
    if name in student_set:
```

```
        print(name, "found")
```

```
    else:
```

```
        print(name, "not found")
```

Output

Charan found

Esha not found

Bhavya found

Task 4 – Refactoring (Replacing Hardcoded Values with Constants)

```
RATE = 2
```

```
TIME = 5
```

```
def simple_interest(principal):  
    interest = (principal * RATE * TIME) / 100  
    print("Simple Interest:", interest)
```

```
simple_interest(1000)  
simple_interest(2000)
```

Output

```
Simple Interest: 100.0  
Simple Interest: 200.0
```

Task 5 – Refactoring (Enhancing Variable Naming and Code Clarity)

```
# Store two numbers  
first_number = 50  
second_number = 30  
  
# Calculate the sum  
total_sum = first_number + second_number  
  
# Display the result  
print(total_sum)
```

Output

```
80
```